THE AHA COMPANY

STELLAR COMMUNITY FUND, ROUND 46, OPEN TRACK

Stratum: multisig for Stellar G and C accounts

A self-custody wallet for groups, with passkeys for signing, policies for permissions and post-quantum keys.

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TEAM

The team behind Stratum

The Aha Company builds it, and engineers join as each deliverable needs them.

WORKING ON IT

PR

Pamphile T. Roy

Lead

Author of Stratum, ghostsig, Skychart and Tansu. Co-author of SEP-52 and SEP-53, SciPy maintainer.

EE

Elizabeth Engelman

Engineer

Took Nido to mainnet with its relayer infrastructure. Built our Fordefi proof of concept.

HH

Hugo Heer

Frontend engineer

Works on the frontends of our products.

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More than 10 engineers on Stellar since before Soroban launched. We built the contract workflows of the Stellar CLI, the Contract Client of the JS SDK, Loam and Scaffold Stellar, and we architected Société Générale-Forge's EURCV. Nido and Authline, our previous SCF awards, are delivered. Others join for Soroban, security reviews or custodian integrations when a deliverable needs them.

AROUND STRATUM

Perch, the contract stack Stratum uses, is ours and is developed by our Stellar Registry team. PaltaLabs contributes to Stratum from outside and is not part of this budget.

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THE PROBLEM

Teams on Stellar share keys across two kinds of account, and no wallet handles both.

Company treasuries, protocol admin keys, DAOs and asset issuers need several people to approve each transaction.

THREE GAPS

01 **G accounts**

Classic multisig counts signatures per type of operation, and nothing more.

02 **C accounts**

Smart accounts enforce rules such as a weekly cap, but their tools serve one owner or one app.

03 **New kinds of keys**

Custodian vaults and post-quantum keys must join the same quorum. Today this is done by hand.

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TRACK RECORD

Stratum is live on mainnet

What exists today, before this award.

PRODUCT

Multisig for G accounts

Payments, contract calls and signer changes, each explained in plain language before anyone signs.

Delivered Live on mainnet since 2025

USERS

PaltaLabs and partners

The team behind Soroswap and DeFindex uses Stratum, and has contributed ten pull requests.

Delivered Contributions since February 2026

CUSTODY

A Fordefi vault as a co-signer

A Fordefi vault signed for a 2-of-2 Stellar account on mainnet, next to a second key.

Delivered Proof of concept, July 2026

PASSKEYS

ghostsig in Stellar Wallets Kit

Our passkey wallet, which any dApp using the kit can offer to its users.

Delivered Since version 2.7.0

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THE SUITE

Self-custody from a portfolio to a group

Four products we build, linked to each other. Users keep their own keys at every step.

01 · Follow

Skychart

A portfolio across Stellar and other chains. It opens any Stellar account in Stratum with one link.

● Delivered

02 · Sign alone

ghostsig

A passkey wallet, in Stellar Wallets Kit since 2.7.0, and already a signer in Stratum.

● Delivered

03 · Sign together

Stratum

Multisig for G accounts today. G and C accounts, policies and post-quantum keys with this award.

○ In progress

04 · Own a smart account

Nido

A passkey smart account on Perch, which joins a Stratum quorum as a member with this award.

● Delivered

Skychart to Stratum

A deep link opens the account in Stratum, on its network.

ghostsig in Stratum

A passkey signs for a G account through Stellar Wallets Kit.

Nido in Stratum

A person's smart account signs as a member of a group's quorum, through CAP-0071.

From watching a portfolio to signing with others, no one else holds the keys, and no one tracks the users.

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WHAT THIS AWARD ADDS

One wallet for every shared account

01	**One key for G and C**	The same Ledger or passkey signs for a G account and is a member of a C account.
02	**Policies**	A Perch policy states who may call what, with how many signatures, under which cap.
03	**Post-quantum members**	An ML-DSA-65 key can be a member of a C account, verified natively since protocol 30.
04	**Passkeys and custodians**	Co-signers join from a link with a passkey, and custodians such as Fordefi join as members.
05	**DeFi from a shared account**	One operator can swap within a cap, with the account as the only recipient.
06	**A SEP draft**	So that any wallet can co-sign the same C accounts.

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ARCHITECTURE

Stratum, its members and the two accounts

A web app with no accounts, no telemetry and no server holding keys, on top of Stellar's two kinds of account.

STRATUM, IN THE BROWSER

G multisig
Explains transactions, refuses lock-outs, warns on flagged addresses.

C policies
Shows the Perch policy, its changes and checks its hash.

Co-signing
Refractor for G and C accounts, links and QR codes.

DeFi
Soroswap and DeFindex calls decoded before signing.

MEMBERS

Ledger
Stellar app 6.1.0 signs CAP-0071 entries.

Passkey
ghostsig on G, the WebAuthn verifier on C.

Custodian vault
Fordefi first, through one interface.

ML-DSA-65 key
On an offline device, answering by QR codes.

STELLAR

G account
Classic multisig: weights and thresholds.

CAP-0071

a G key is a member

C account (Perch)
Policy document, verifiers, recovery controller, spending cap.

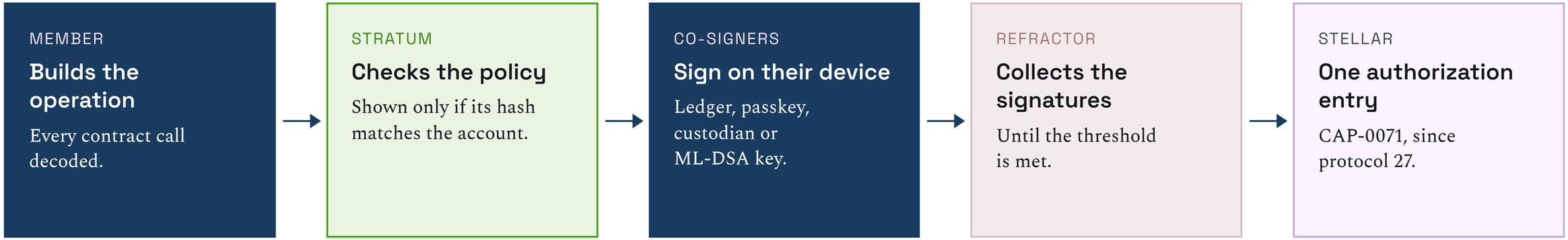
Stratum adds no contract. The C account runs Perch's stack, deployed on testnet and as a beta on mainnet.

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HOW IT WORKS

One signing session for G and C accounts

Each approval covers one exact operation, and every device checks the policy before signing.



What exists

Stratum for G accounts, and Perch's contracts on testnet and as a beta on mainnet.

What we build

Collecting signatures for C accounts, the policy view, onboarding and recovery screens.

What we do not build

New contracts. Stratum uses Perch's, and never holds anyone's keys.

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DEPENDENCIES

What Stratum builds on

As of 7 November 2026, from the public repositories and release notes.

Component	What it gives Stratum	Owner	Status
Perch	Policies, the account contract, passkey verifier and recovery. Deployed on testnet and as a beta on mainnet, audit scoped.	The Aha Company	In progress
ghostsig	A G key derived from a passkey, with Shamir backups.	The Aha Company	Delivered
Ledger Stellar app 6.1.0	Signs CAP-0071 authorization entries and shows the account they are bound to.	Ledger	Third party
Protocol 30	Native ML-DSA verification (CAP-0087) and muxed contract addresses (CAP-0084).	Stellar validators	Third party
Refractor	Collects signatures until the threshold is met. C account authorization entries are in its Public Goods plan.	StellarExpert	Third party

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OPEN QUESTIONS

What we will work out while building

Part of this award is the time to settle these well, in the open and with the teams that use Stratum.

01	**Approvals between wallets**	How a half-signed authorization entry moves from one wallet to another. We work it out in a public SEP discussion.
02	**Approvals and policy changes**	Perch binds a policy change to its revision, not yet an ordinary approval. How should a wallet learn which rules an account follows?
03	**Custodians on C accounts**	Whether a custodian such as Fordefi can sign for a delegated member, or stays a G-account signer.
04	**Recovery for a group**	How a quorum chooses between keeping the account active and freezing it while a recovery runs.
05	**DeFi operators**	How narrow an operator rule can be while a swap still goes through.
06	**Smart accounts as members**	How a person’s own smart account, such as a Nido wallet, signs inside a group’s quorum.

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Thank you

stellar-stratum.xyz
github.com/tupui/stellar-stratum

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